Ebenezer Tarubinga

AI/ML ENGINEER · SEOUL, KOREA

ebstarmusic@gmail.com · psychofict.github.io · github.com/psychofict · ORCID 0009-0004-7340-1873 · in/ebstar

First-author researcher in semi-supervised semantic segmentation — training dense-prediction models from very few labels without letting their own mistakes compound. Four papers, two peer-reviewed. In production: retrieval, agents and on-device vision for smart-city infrastructure.

PUBLICATIONS — FIRST AUTHOR

01 **CW-BASS v2: Saturation-Aware Pseudo-Label Selection for Semi-Supervised Segmentation under Foundation-Model Teachers**
Under review, **IEEE TPAMI** · 2026
arXiv:2608.12773 · 2nd among DINOv2-backbone methods · +1.5 mIoU on ADE20K

02 **PixCon: Clean-Positive Contrastive Learning for Foundation-Model Semi-Supervised Segmentation**
Under review, **CVPR 2027** · 2026
arXiv:2607.03068 · #2 global SSSS SOTA · matches UniMatch V2 at lower cost

03 **FARCLUSS: Fuzzy Adaptive Rebalancing and Contrastive Uncertainty Learning for Semi-Supervised Semantic Segmentation**
Neural Networks 205 (2027) 109494 · Q1, top 10% impact factor · with J. Kalafatovich, S.-W. Lee
doi:10.1016/j.neunet.2026.109494 · online 13 Aug 2026 · former #2 global SSSS SOTA · 78.8 / 78.2 mIoU on Cityscapes & Pascal VOC, ResNet-101

04 **CW-BASS: Confidence-Weighted Boundary-Aware Learning for Semi-Supervised Semantic Segmentation**
IEEE IJCNN 2025 · CORE A · with J. Kalafatovich, S.-W. Lee
arXiv:2502.15152 · former #3 global SSSS SOTA · 77.15 mIoU on Pascal VOC 2012, ResNet-50

PATENT

Real-time object detection method, recording medium and device performing the same — Application filed
Republic of Korea
Filed out of the autonomous-driving perception work at Korea University.

EXPERIENCE

AI/ML Engineer
Sept 2025 — present
Gractor · Seoul, Korea

- Design and ship production retrieval (RAG) and tool-calling agent systems for smart-city platforms, with guardrails at the tool boundary and evaluation harnesses that cover adversarial inputs.
- Develop and optimise computer-vision models for urban sensing, deploying on-device inference to municipal edge hardware.
- Integrate inference services into live systems with cross-functional teams, improving reliability and latency.

Founder
2026 — present
Ebenworks · Seoul, Korea

- AI product company. Lead product and engineering direction, turning applied AI research into shipped software.

Research Engineer (MSc)
Aug 2023 — Feb 2026
Pattern Recognition & Machine Learning Lab, Korea University · Seoul, Korea

- Four first-author papers in semi-supervised semantic segmentation, advised by Prof. Seong-Whan Lee (IEEE Fellow).
- Built roughly 10K lines of multi-GPU PyTorch training and evaluation infrastructure.
- Object detection and tracking for autonomous driving; one Korean patent filed.
- Further work in monocular depth estimation, dense and sparse matching, and instance retrieval.

AI Software Engineer

Jan 2019 — Jan 2021

GliT · Hybrid

- Built offline-first mobile learning products reaching 500+ students and 80,000+ learning sessions.
- Delivered 10+ full-cycle product builds end to end, led the startup's technical direction, and ran STEM programmes with 10+ partner schools.

EDUCATION

MSc, Artificial Intelligence

2023 — 2026

Korea University · Seoul, Korea

Computer vision, advised by Prof. Seong-Whan Lee (IEEE Fellow). **Global Korea Scholarship** — sole awardee from Zimbabwe, 2022. **BK21 Research Fellowship.**

TECHNICAL SKILLS

AI / LLM	RAG (hybrid retrieval), agents & tool-calling, Model Context Protocol, evaluation harnesses, guardrails, LangGraph, OpenAI, Anthropic Claude, HyperCLOVA X, Hugging Face
ML / VISION	PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, YOLOv5, OpenVINO, ONNX, CUDA, Weights & Biases
BACKEND & DATA	FastAPI, Flask, Node.js, Express, SQLAlchemy, Prisma, knowledge graphs, PostgreSQL + PostGIS, MongoDB, Redis, OpenSearch
FRONTEND	React 19, TypeScript, Next.js, Three.js, deck.gl, MapLibre, ECharts, Tailwind, Vite
INFRA & IOT	Docker, Kubernetes, AWS, Nginx, Prometheus, Grafana, Mosquitto MQTT, ModBus RTU, Vercel, GitHub Actions
LANGUAGES	Python, TypeScript, JavaScript, C++, C#, Java, SQL, Bash

SELECTED OPEN SOURCE & RESEARCH CODE

hwpkit

github.com/psychofict/hwpkit

Python library to read, fill and edit Korean HWP (Hancom Office) documents — text extraction for LLM and RAG pipelines, programmatic form-filling, and a binary rewrite that does not corrupt the file.

Research studies

github.com/psychofict

Backdoor attacks on CLIP; semantic-aware multi-label adversarial attacks; semi-supervised segmentation baselines (UniMatch, ST++); monocular depth estimation on KITTI; effective context length of LLMs (STRING). Trained checkpoints at huggingface.co/psychofict.

LEADERSHIP

Founder & CEO — The ESØTËRIC Ones

2023 — present

Independent record label: 14 artists across six countries, 30+ releases, 5M+ streams. Built the distribution and analytics tooling, and automated royalty tracking and reporting.

Director of Public Relations — Zimbabwean Community in South Korea

Apr 2025 — present

Inaugural board member. Built the community's digital strategy and ran technology-focused career and education programmes for its youth.

GINCON Committee Member — Korean National Assembly

2025

One of several cultural and technology diplomacy roles with Korean national and provincial bodies, including the Ministry of Foreign Affairs and the Seoul Tourism Organisation.